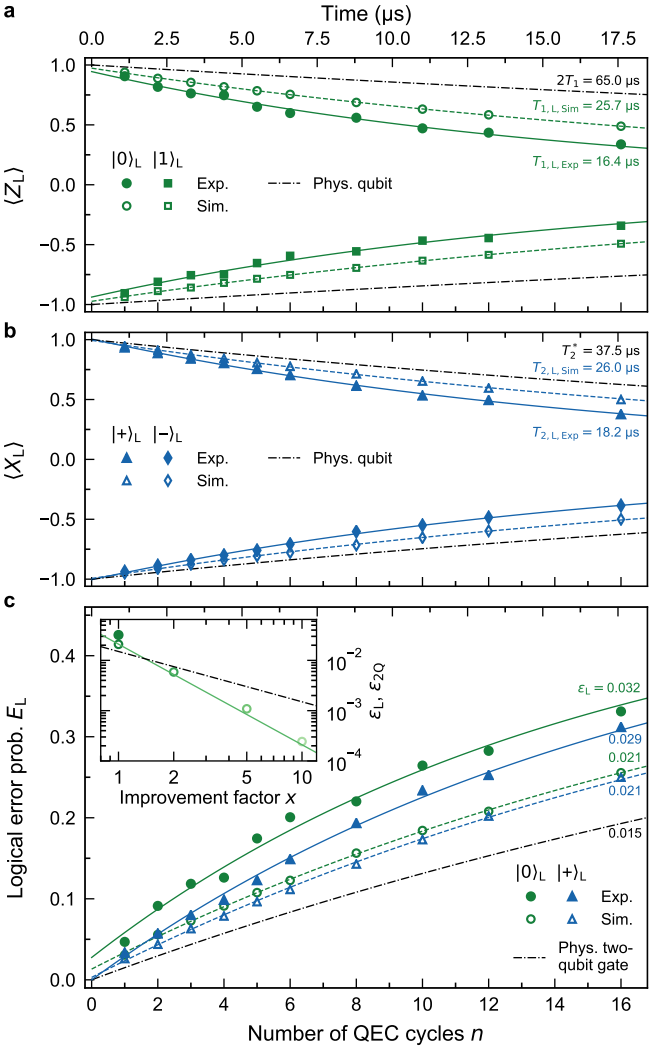
a
Time (μs)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
1.0
0.5
0.0
−0.5
−1.0
⟨Z_L⟩
2T_1 = 65.0 μs
T_1, L, Sim = 25.7 μs
T_1, L, Exp = 16.4 μs
|0⟩_L |1⟩_L
Exp.
Sim.
Phys. qubit
b
⟨X_L⟩
T_2^* = 37.5 μs
T_2, L, Sim = 26.0 μs
T_2, L, Exp = 18.2 μs
|+⟩_L |−⟩_L
Exp.
Sim.
Phys. qubit
c
Logical error prob. E_L
0.4
0.3
0.2
0.1
0.0
0
2
4
6
8
10
12
14
16
Number of QEC cycles n
ε_L = 0.032
0.029
0.021
0.021
0.015
|0⟩_L |+⟩_L
Exp.
Sim.
Phys. two-qubit gate
Improvement factor x
1
2
5
10
10^−2
10^−3
10^−4
ε_L, ε_2Q